| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2010 | **Task Last Updated:** | FY 08/16/2010 |
| **PI Name:** | Ploutz-Snyder, Lori L. Ph.D. | | |
| **Project Title:** | Evaluation of Maximal Oxygen Consumption (VO2pk) Protocols for Determining Ventilatory Threshold | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Biomedical countermeasures | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | lorips@umich.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | (734) 764-5210 |
| **Organization Name:** | University of Michigan | | |
| **PI Address 1:** | OBL 4170, 1402 Washington Hts. | | |
| **PI Address 2:** | School of Kinesiology | | |
| **PI Web Page:** | | | |
| **City:** | Ann Arbor | **State:** | MI |
| **Zip Code:** | 48109-2013 | **Congressional District:** | 12 |
| **Comments:** | Previously at Universities Space Research Association/NASA Johnson Space Center until July 2016. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | Directed Research |
| **Start Date:** | 11/13/2009 | **End Date:** | 04/30/2010 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Norsk, Peter | **Contact Phone:** | |
| **Contact Email:** | Peter.norsk@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Everett, Meghan ( University of Houston )<br>Guined, Jamie ( University of Houston )<br>Redd, Elizabeth ( University of Houston ) | | |
| **Grant/Contract No.:** | Directed Research | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | There are many different cycle protocols that can be used to elicit maximal oxygen consumption or VO2pk. They all involve incremental increases in workload until maximal exercise capacity is reached. It is generally accepted that ideal protocols should include multiple gradual increases in workload (as opposed to only a few large increases in workload) with a total test time of around 8-14 minutes. Our “standard” cycle protocol involves only 3 step increases in workload between 0-75% of max, with smaller increments thereafter. We are interested in refining this protocol to allow for small increments in workload throughout the entire test. This would allow us to gather more information about subject fitness, such as ventilatory threshold as well as be more comfortable for the subjects. Ventilatory threshold has been shown to be more predictive of a subject’s ability to work submaximally over a longer time whereas VO2pk assesses the maximal ability. Before implementing any new protocols in research studies we would like to first try them in the lab. The primary | | |

| | |
|---|---|
| **Task Description:** | purpose of this study is to assess whether ventilatory threshold can be determined in four different maximal aerobic exercise protocols and to assess subject comfort. The study protocols differ in workload and stage time.<br>Our standard VO2pk measurements will be obtained. During the test subjects will breathe in normal air but their exhaled breath will be collected to measure exercise oxygen consumption using the same metabolic gas analysis system (ParvoMedics) employed by the ExPC Laboratory for astronauts and bed rest subject. To accomplish this, subjects will breathe through a mouthpiece similar to a snorkel, and their nose will be occluded with a nose clip. Subject's blood pressure will be monitored with a stethoscope and blood pressure cuff before exercise begins and every two minutes during the exercise protocol and recovery period, heart rate will be monitored continuously via 3-lead electrocardiography (ECG), and rating of perceived exertion will recorded each exercise stage using a standardized scale. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | Exercise Protocols. Subjects were asked to complete 2-4 cycle ergometer tests to peak exertion. Each protocol consisted of a three minute warm-up stage, followed by either 30 second or 60 second stages at increasing workloads until peak exertion. Selection of the protocols was based on the subject's weight. Subjects weighing less than 65 kg completed protocols A and B, and subjects weighing more than 65 kg completed protocols C and D.<br>Protocol A: The exercise test session will include three minutes of warm-up at 0 watts followed by 60 second stages, increasing 15 watts each stage until peak exertion.<br>Protocol B: The exercise test session will include three minutes of warm-up at 0 watts followed by 30 second stages, increasing 7.5 watts each stage until peak exertion.<br>Protocol C: The exercise test session will include three minutes of warm-up at 50 watts followed by 60 second stages, increasing 25 watts each stage until peak exertion.<br>Protocol D: The exercise test session will include three minutes of warm-up at 50 watts followed by 30 second stages, increasing 12.5 watts each stage until peak exertion.<br>Taking the results together with the subject preference for 60 second stages, it is the recommendation that future bed rest and flight studies implement Protocol A for individuals that are less than 65 kg and demonstrate lower levels of fitness; and Protocol C for individuals that are above 65 kg and exhibit higher levels of fitness. Determination of VT can be achieved in peak cycle tests as defined by these protocols, and by implementing the data analysis methods used in this study. |
| **Bibliography Type:** | Description: (Last Updated: 11/21/2025) |